

Assessment of AK

Presentation of AK

Clinically, actinic keratoses (AKs) typically present as scaly or keratotic patches, papules, or plaques on an erythematous base. Palpation reveals a sandpaper-like texture. The diameter usually does not exceed 1 cm, although in some patients, lesions can be numerous and confluent. Lesions are often skin-colored but may also appear pink, red, or brownish. The surrounding skin frequently shows signs of chronic sun damage, including telangiectasias, dyschromia, elastosis, and wrinkles.

Depending on their clinical and histological features, several variants of AK have been described, such as pigmented, atrophic, Bowenoid, lichenoid, and hyperkeratotic types.

The anatomical distribution of AKs correlates strongly with sun-exposed areas, underscoring the role of ultraviolet radiation in their development.

Clinical Diagnosis

Table 3 outlines the expert panel's recommendations for the assessment of AK lesions.

Histological Definition and Assessment of AK

The primary histological criterion for classifying the severity of AK, as proposed by Röwert-Huber (2007) and Cockerell (2000), is the extent of atypical keratinocytes within the epidermis. This classification is summarized in Table 4.

Subgroups of Patients Presenting with AK

There is no universally accepted definition for grading the overall severity of AK. However, distinct patient subgroups with differing therapeutic needs were defined during the development of these guidelines to reflect real-world clinical practice. These subgroups were discussed and agreed upon during the initial consensus conference (see Table 5).

Table 5: Recommendations for a Classification of Patients According to the Severity of Actinic Keratosis (AK)

Recommendations for a classification of patient subgroups	Evidence	Percentage of agreement
The following subgroups of patients should be considered separately: (1) Single AK lesions: At least one and no more than five palpable or visible AK lesions per field or affected body region (2) Multiple AK lesions: At least six distinguishable AK lesions in one body region or field (3) Field cancerization: At least six AK lesions in one body region or field, with contiguous areas of chronic actinic sun damage and hyperkeratosis (4) Immunosuppressed patients with AK: AK at any severity level and concomitant immunosuppression (e.g., due to chronic immunosuppressive medication or diseases affecting immune function, such as hematological malignancies)	Expert consensus	≥90%

Treatment Options

The following interventions were selected as relevant for the treatment of AK, based on consensus by at least 75% of the expert panel. These treatments and their specified modes of application formed the inclusion criteria for the systematic literature review.

Interventions and application methods not listed were excluded from the review; however, this does not imply they are ineffective or inappropriate. When applying these guidelines in clinical practice, healthcare providers should adapt treatment modalities according to national regulations and individual patient factors. Particular attention should be paid to approval status, dosage, dosing regimen, adverse effects, contraindications, and potential drug interactions.